REZON fall detector · parts list

Hackathon build with the SM-24 geophone and a SIM868 call alert · casing 3D-printed on site · no breadboard

2026-11-12
Row = line in the Hestore stock CSV

SIGNAL CHAIN

SM-24 geophone → ADC 15 Click → SPI → ESP32-S3
ESP32-S3 → I²S → PCM5102A → PAM8403 → Speaker
ESP32-S3 → UART → SIM868 (Waveshare) → GSM call → our phone

Plus a WS2812B status LED (green / amber / red) and an “I’m OK” button. One 18650 cell feeds the SIM868 directly and a 5 V boost for everything else.

OURS / TO BUY (NOT IN HACKATHON STOCK)

- **SM-24 geophone** (here)
- **MikroE ADC 15 Click** (ADS131M02, already running on the bench)
- **Waveshare Pico-SIM868 board** (here) + its **GSM antenna**: check the box, buy one if missing
- **Nano SIM** from one of our phones, PIN off
- **TP4056 USB-C charger module** with DW01 protection
- **To buy:** PH2.0 2-pin plug with leads (battery lead into the SIM868 board)

SIM868 SETUP

- **Pull the Pico H off** the SIM868 board: it is not used. The ESP32-S3 drives the board through 5 M-F jumpers into its Pico sockets: **pin 1** RXD ← GPIO17, **pin 2** TXD → GPIO18, **pin 3** GND, **pin 19** PWRKEY ← GPIO16, **pin 36** 3V3 ← ESP32 3V3.
- Power it **straight from the 18650** (3.4–4.4 V) into the board’s **PH2.0 battery socket**, its power slide switch on the battery side. Never 5 V; header pins 39 (VSYS) and 40 (VBUS) stay empty.
- Bursts up to 2 A while transmitting: **1000 µF** at the take-off on the perfboard, short 0.5 mm² leads.
- Logic: the board has its own 3.3 V ↔ 2.8 V level shifter (fed from pin 36), so **no divider** any more.
- Power-on: **GPIO16 HIGH for 1.5 s** (an NPN pulls PWRKEY low), then wait for **RDY**. NET LED: fast blink = searching, about every 3 s = registered.
- **GSM antenna** on the “GSM” u.FL socket before the first power-on. GNSS and BT antennas are not needed.
- SIM868 is **2G only**: Hungary still runs 2G (Telekom’s 2G now runs on Yettel’s network); confirm coverage at the venue.
- Turn the SIM PIN off in a phone first; test **AT** → **AT+CPIN?** → **AT+CREG?** and a call before the show.

Sensing & processing

	QTY	PART	ROW	SOURCE	PURPOSE
☐	1	SM-24 geophone (SparkFun SEN-11744)	–	Ours	Floor vibration sensor, 28.8 V/(m/s), 10–240 Hz
☐	1	MikroE ADC 15 Click (ADS131M02, 24-bit, 2 ch) MIKROE-4890	–	Ours	Geophone straight into the ADC at gain 32 (±37.5 mV full scale); firmware already brought up
☐	1	ESP32 S3 fejlesztői modul, 16MB Flash, 8MB PSRAM, WiFi, BT5, USB-C ESP32-S3-WROOM-1-N16R8-M	179	Hackathon	Main controller; exact target of our firmware (PSRAM for buffers and the model). Also drives the SIM868 board
☐	1	16 bites analog-digital konverter modul, I2C BACKUP ADS1115-M	11	Hackathon	Only if the Click fails: ±0.256 V range, max 860 SPS; enough for fall impacts

Alert: call, voice, light, button

	QTY	PART	ROW	SOURCE	PURPOSE
☐	1	Waveshare Pico-SIM868-GSM/GPRS/GNSS board (SIM868: 2G GSM/GPRS + GNSS + BT)	–	Ours	Calls our phone when a fall is confirmed and nobody presses “I’m OK”. Used as a modem only: Pico H removed, the ESP32-S3 drives it over UART
☐	1	GSM antenna with u.FL (IPEX) plug, 850/900/1800/1900 MHz	–	Ours	Goes on the board’s “GSM” socket; mandatory. It should be in the Waveshare box (none is fitted in our photo): buy one if it is missing
☐	1	PH2.0 (JST-PH 2.0 mm) 2-pin plug with ~10 cm leads	–	Buy	18650 VBAT into the SIM868 board’s battery socket. Not in the hackathon stock; check polarity against the box’s Li-po plug
☐	1	SIM card from one of our phones (PIN disabled), nano	–	Ours	Network access for the call; must work on 2G. The SIM868 board takes a nano SIM directly, no adapter
☐	1	PCM5102A alapú DAC audió dekóder I2S modul	249	Hackathon	Audio out for “Mari néni, jól van?”; the S3 has no built-in DAC
☐	1	Hangszóró erősítő, PAM8403 alapú 2x3W erősítő modul	269	Hackathon	Speaker amplifier, runs from the 5 V rail
☐	1	Hangszóró, kicsi 1W 8ohm	262	Hackathon	Speaker
☐	1	WS2812B RGB vezérelhető LED szalag, 1 m, 60 LED/m WS2812B-LS-60-IP20-BK	68	Hackathon	Cut one pixel: green / amber / red status LED
☐	1	1N4001 dióda	364	Hackathon	Drops the LED supply to ~4.3 V so it reliably reads the 3.3 V data line

	QTY	PART	ROW	SOURCE	PURPOSE
☐	**2**	TACT mikrokapcsoló, SPST-NO, THT, 6x6mm, 5mm `TACT-65R-F`	51	**Hackathon**	“I’m OK” / cancel-alarm button (+1 spare)

Power

	QTY	PART	ROW	SOURCE	PURPOSE
☐	**2**	Li-Ion akkumulátor, 18650, 3.6V, 3500mAh `INR18650-MJ1`	2	**Hackathon**	Battery (+1 spare); handles the SIM868’s 2 A bursts easily
☐	**1**	Elem / akkumulátortartó, 1 x 18650, vezetékkel `1 X 18650 W`	3	**Hackathon**	Battery holder
☐	**1**	Világítás billenőkapcsoló, 12V, SPST, OFF-ON `AK-01`	46	**Hackathon**	Main power switch; carries the 2 A bursts (the small slide switch is only rated 0.5 A)
☐	**1**	DC-DC boost	304	**Hackathon**	Cell → 5 V rail for the ESP32 5V pin, amp, DAC and LED only (not the SIM868); set to 5.0 V
☐	**1**	TP4056 USB-C charger module with DW01 protection	–	**Ours**	Battery charging; no charger on the hackathon list
☐	**1**	Fémréteg ellenállás készlet, 500 db, 50 érték, 0.5W, 1% `RESISTOR-KIT-500P`	165	**Hackathon**	1 kΩ across the geophone coil · 2 × 100 kΩ battery divider · 10 kΩ button pull-up (no SIM divider any more)
☐	**1**	Elektrolit kondenzátor készlet, 0.1uF–1000uF, 500 db `ELCOPACK-500`	166	**Hackathon**	1000 µF at the SIM868 power take-off · 470–1000 µF on the 5 V rail
☐	**1**	Kerámia kondenzátor készlet, 24 féle, 960 db `CER-CAP-PACK`	167	**Hackathon**	100 nF decoupling at each module’s supply

Wiring & assembly

	QTY	PART	ROW	SOURCE	PURPOSE
☐	**1**	9x15 cm, kétoldalas furatgalvánozott, univerzális nyák, FR4 `DPCB-915`	101	**Hackathon**	Base board (cut to fit the case); the SIM868 board stands off-board at the back wall
☐	**2**	Tüskesor aljzat, egyenes, 40 pin `ZL262-40SG`	131	**Hackathon**	Sockets so the ESP32-S3 and Click stay swappable
☐	**1**	Tüskesor, egyenes, tördelhető, 40 pin `ZL201-40G (WWS40G)`	130	**Hackathon**	Pin headers for modules that lack them, and H2 (5 pins) for the SIM868 jumpers
☐	**1**	Szalagkábel csatlakozóval, 10cm, 40p, female-female (jumper) `RC-40-10/FF`	42	**Hackathon**	Click mikroBUS → ESP32-S3 (SCK, SDO, SDI, CS1, CS2, INT, RST, 3V3, GND), lid harness
☐	**1**	Szalagkábel csatlakozóval, 10cm, 40p, male-female (jumper) `RC-40-10/MF`	39	**Hackathon**	5 jumpers H2 → SIM868 board: the male end goes straight into the board’s Pico sockets
☐	**1**	kábel 6x0,22	359	**Hackathon**	Geophone lead (one twisted pair), about 1 m; board links
☐	**1**	Vezeték, H05V-K, sodrat, Cu, 1 x 0.5mm², piros	353	**Hackathon**	Battery + / 5 V / SIM868 battery lead
☐	**1**	Vezeték, H05V-K, sodrat, Cu, 1 x 0.5mm², fekete	352	**Hackathon**	GND
☐	**1**	Egymagos vörösréz huzal 20 AWG, 15 m	349	**Hackathon**	Rails and buses on the perfboard
☐	**1**	Zsugorcső készlet (2:1), 530 db `HST-SET530`	157	**Hackathon**	Insulation
☐	**1**	USB 2.0 kábel, USB A – USB C, 3A, 1.5m `CTHBG`	138	**Hackathon**	Flashing and serial monitor
☐	**1**	M3 távtartó készlet	205	**Hackathon**	Mount the board in the printed case
☐	**1**	Csavar készlet M2–M6	302	**Hackathon**	Fixings
☐	**1**	Kábelkötegelő (100 db), fekete, 2.5 × 100 mm `CV 100/2,5/B`	64	**Hackathon**	Cable management and strain relief

Second SM-24, if it turns up: same wiring on the Click’s CH2, also with a 1 kΩ resistor across the coil.

Came with the SIM868 but not used: the Raspberry Pi Pico H, the 600 mAh Li-po (one 18650 feeds everything), the GNSS and BT antennas. Its small speaker is optional (SIM868 speaker output for two-way talk during the call). Left out on purpose: casing, breadboards, soldering tools, displays, other accelerometers.